Distinctive Features

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Phonemes - Contrast

p ≠ b

- pin
- nap
- tripe
- bin
- nab
- tribe

Phonemes - Contrast

l ≠ p

- lick
- pipe
- pine
- pick
- pile
- line

Phonemes Contrast

t ≠ d

- time
- bit
- tome
- dime
- bid

- dome

Phonemes - Contrast

Is the *way* that

p ≠ b

more similar to how

l ≠ p

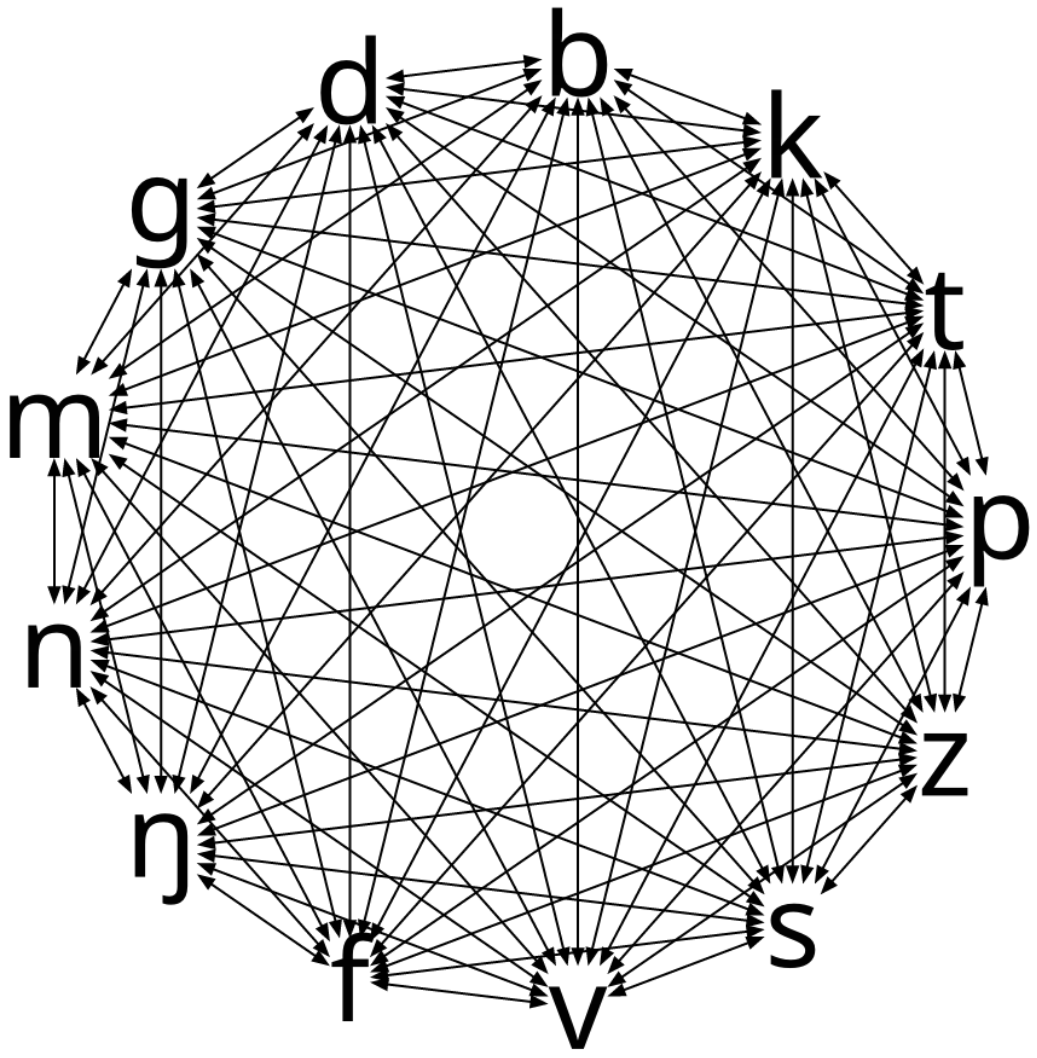
or to to how

t ≠ d

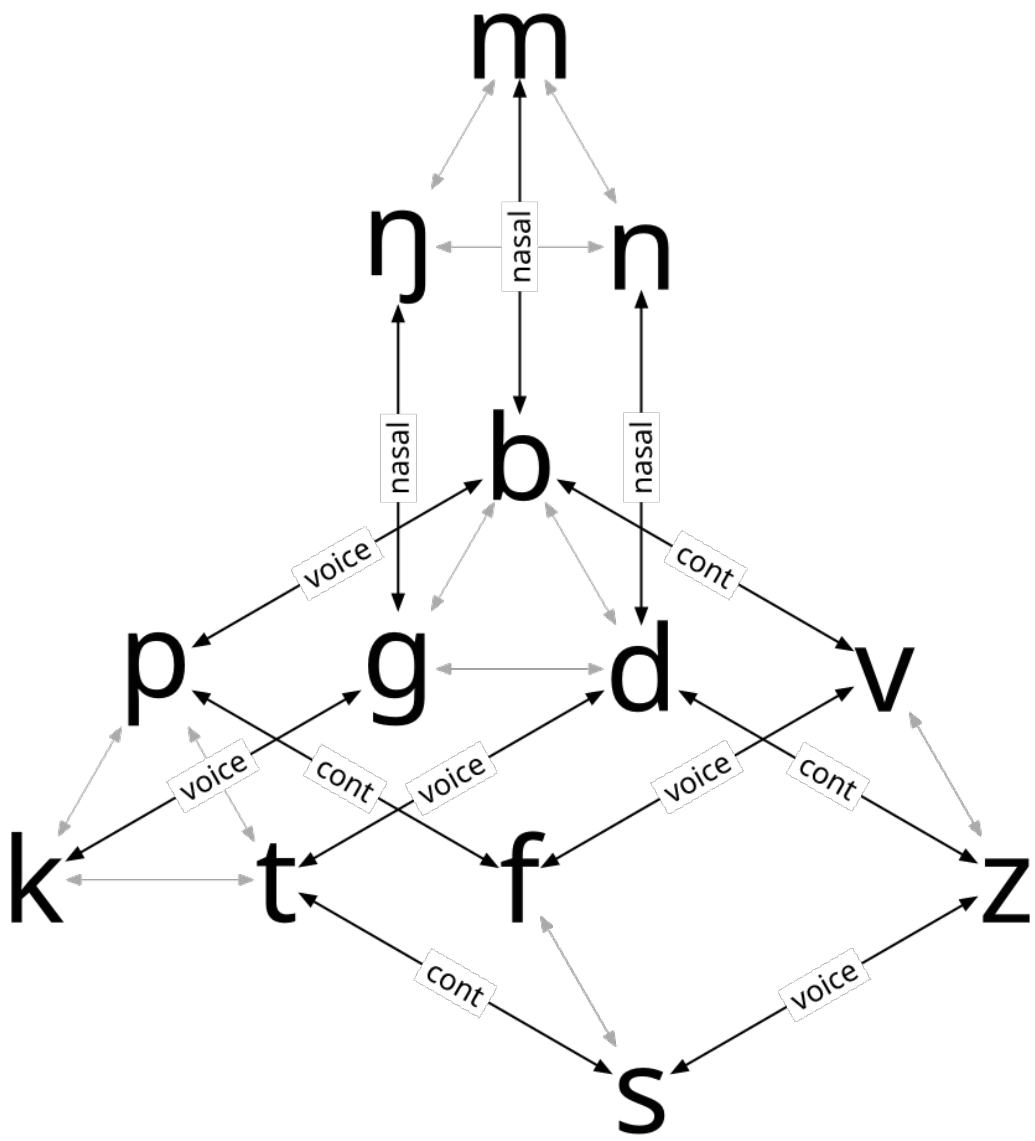
?

Phonemes - Contrast

If all contrasts are the same, then...

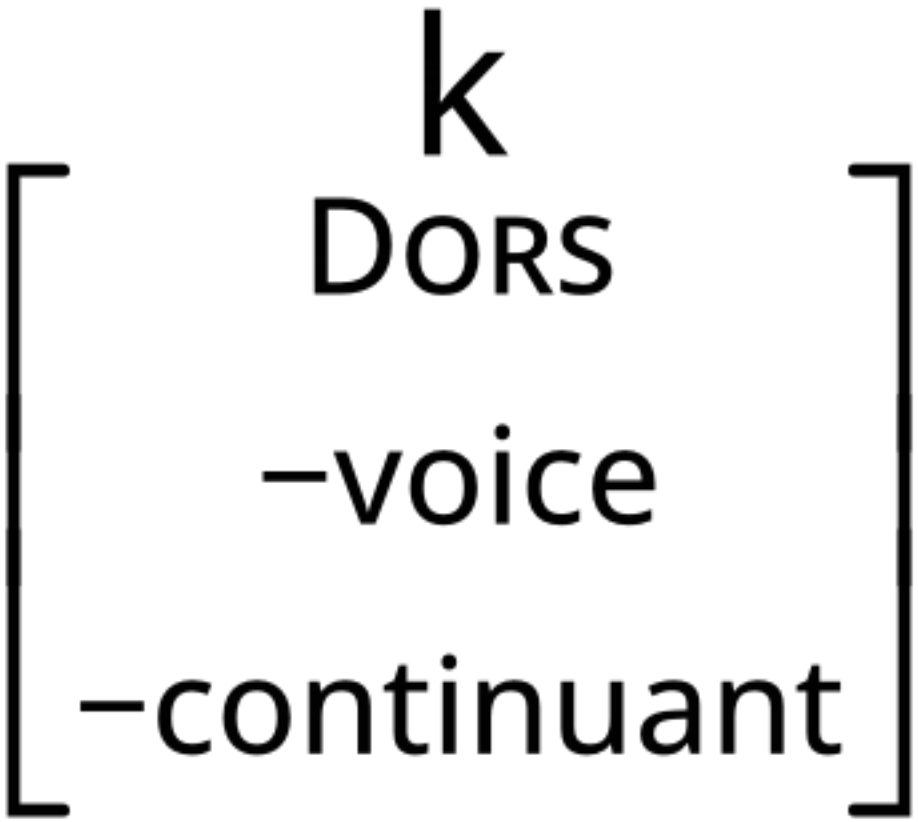

b
k
t
p
z
s
v
f
ŋ
n
m
g
d

Distinctive Features



Distinctive Features

Each phoneme is represented as the set of distinctive features that distinguishes it from all other phonemes



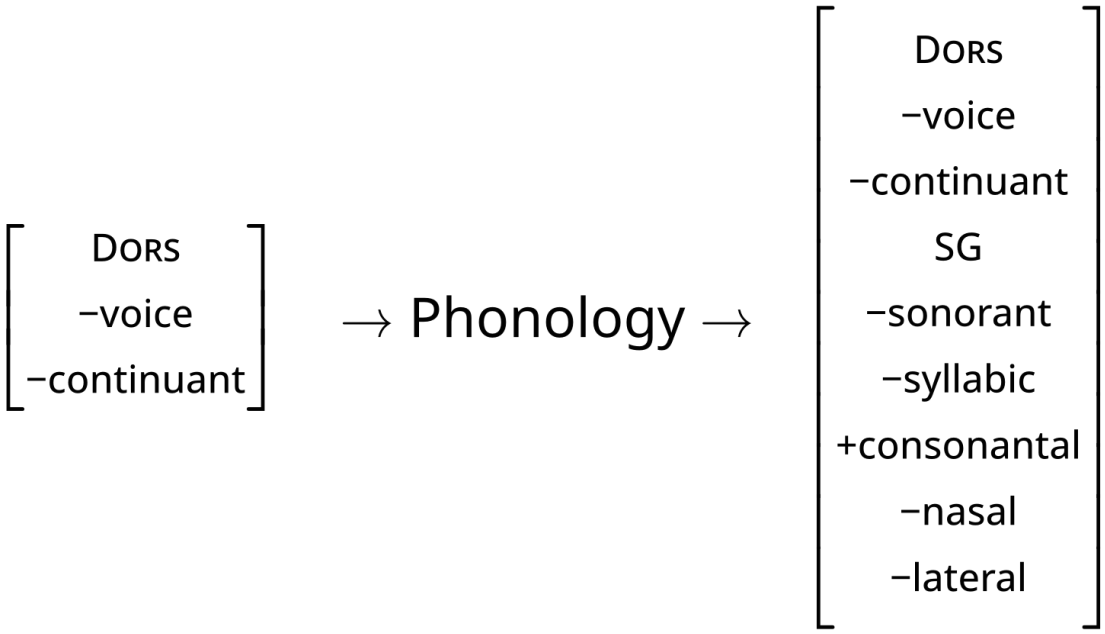
Distinctive Features

A “fully specified” phone has every possible feature specified

$$k^h$$

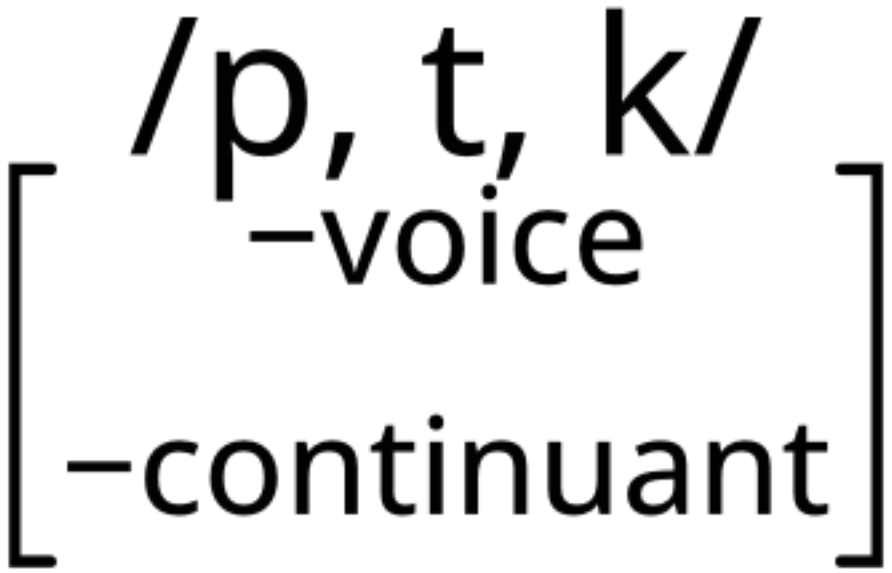
$$\begin{bmatrix} \text{DORS} \\ -\text{voice} \\ -\text{continuant} \\ \text{SG} \\ -\text{sonorant} \\ -\text{syllabic} \\ +\text{consonantal} \\ -\text{nasal} \\ -\text{lateral} \end{bmatrix}$$

Distinctive Features



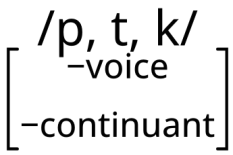
Distinctive Features

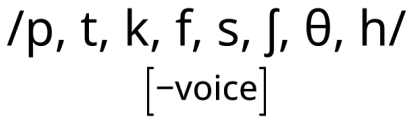
A set of distinctive features describes a natural class



Distinctive features

The fewer the features, the larger the class



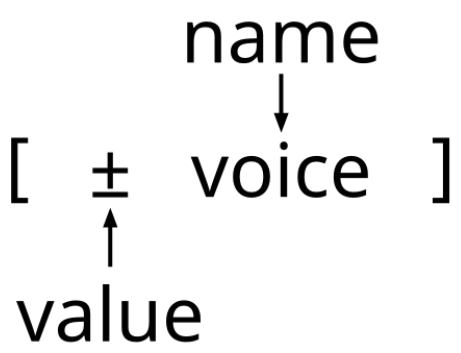


Distinctive Features - Universal?

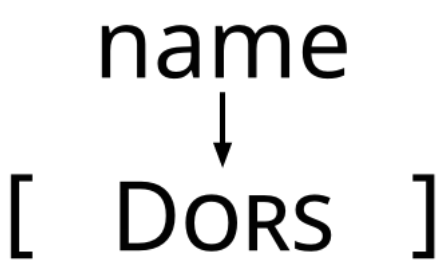
There are conventionalized names for most distinctive features, and some researchers suggest there is a universal set of features that each language can draw from.

Distinctive Features - Notation

“Binary” features



“Monovalent” features



Major Class

[± syllabic]

[+] = Can be nucleus of a syllable.

[± consonantal]

[-] = No significant oral tract constrictions. (Often shortened to “V” and “C”)

Major Class

	[-syllabic]	[+syllabic]
[+consonantal]	Your typical C	[bʌʔn̩, wɔtɻ̩]
[-consonantal]	[jɛs]	Your typical V

Manner

[±sonorant]

[-] = build up of pressure behind oral constriction

[±continuant]

[-] = complete closure

[±nasal]

[+] = open velum

[±lateral]

[+] = closure down the center, open to the sides

Manner

	[+sonorant]	[-sonorant]
[+continuant]	glides, approximants	fricatives
[-continuant]	nasal stops	oral stops

Laryngeal Features

[±voice]

[+] = vocal fold vibration

[SpreadGlottis]

Large glottal opening

[ConstrictedGlottis]

Constricted vocal folds

Laryngeal Features

		[SG]
[+voice]	[b, d, g]	[bʱ, dʱ, gʱ] (breathy)
[-voice]	[p, t, k]	[pʰ, tʰ, kʰ]

“Laryngeal Realism”

Some people argue that [±voice] is actually [Voice].

English

Features	Segments
[]	/b, d, g/
[SG]	/p, t, k/

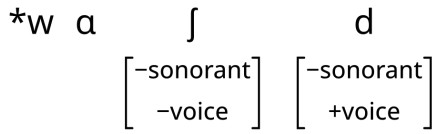
Dutch

Features	Phonemes
[Voice]	/b, d, g/
[]	/p, t, k/

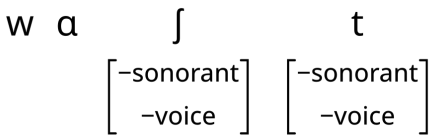
Underspecification

Not every phoneme that has phonetic voicing is necessarily [+voice]. e.g. anything [+sonorant] might get voicing “for free.”

Underspecification

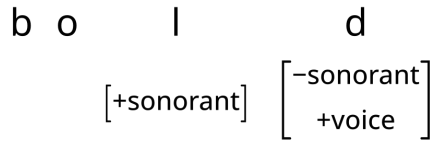


washed

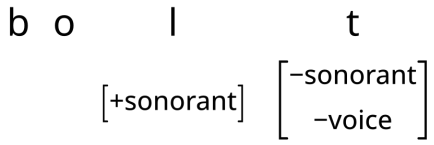


washed

Underspecification



bold



bolt

Major Place Features

All privative or monovalent features

[Labial]

Involving the lips.

[Coronal][1]

Front part of the tongue

[Dorsal][2]

Back part of the tongue

[Pharyngeal]

Involving the pharynx

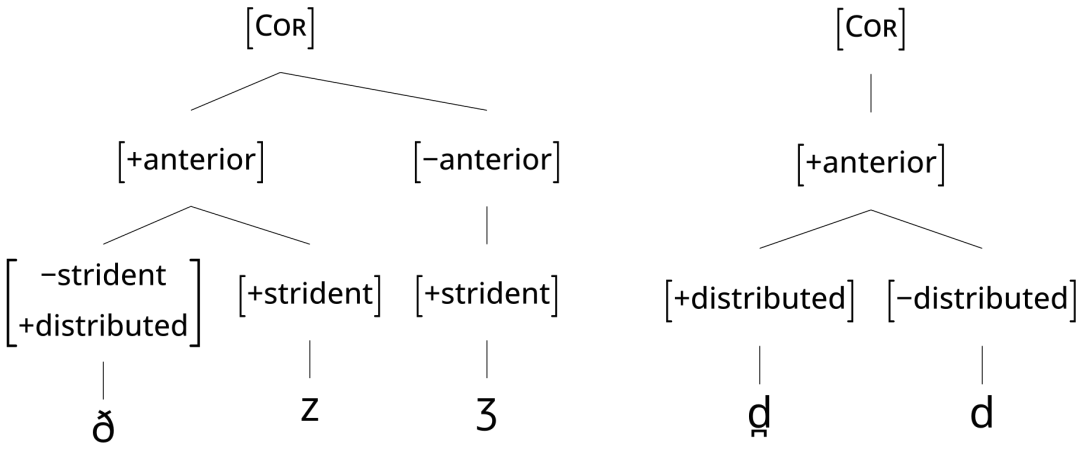
[Laryngeal]

Constriction at the glottis

[1]Not to be confused with the “coronal plane” from articulatory imaging!
[2]Not to be confused with the “dorsal plane” from articulatory imaging!

Subsidary Place Features

The broad strokes place features can't capture all contrastive places differences

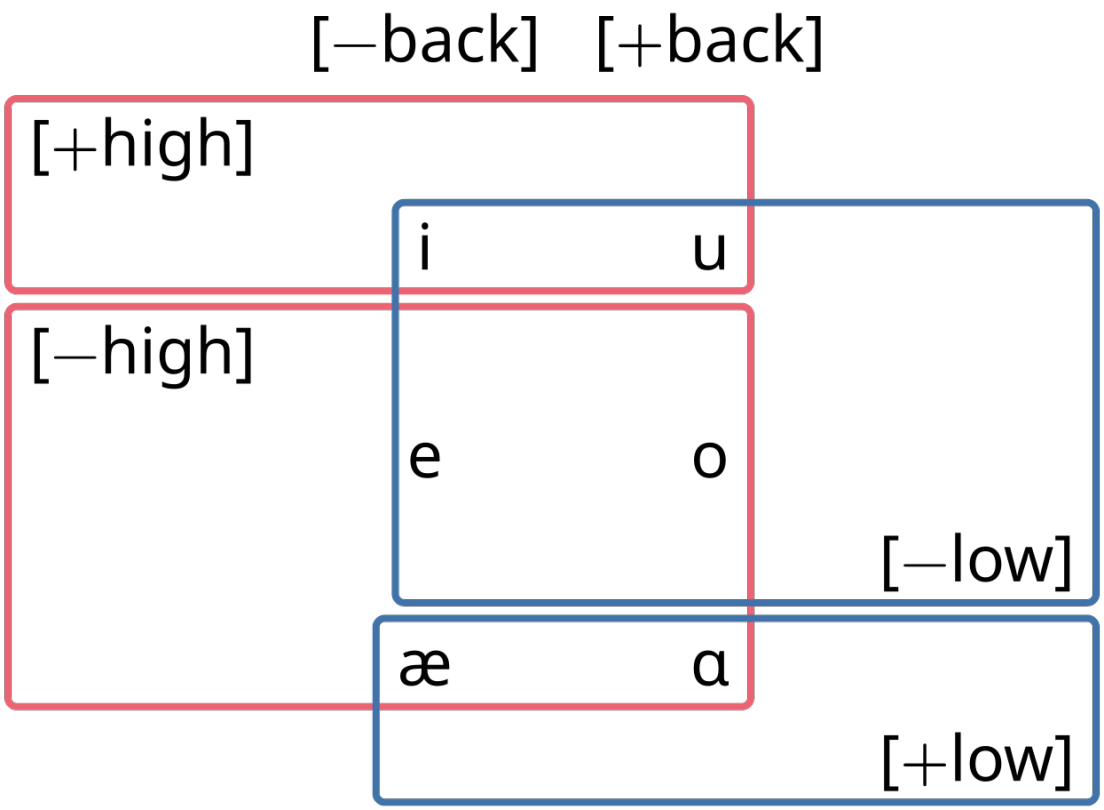


Vowels

Vowels are all going to be

$$\begin{bmatrix} \text{Dors} \\ +\text{syllabic} \\ -\text{consonantal} \end{bmatrix}$$

Vowels



Vowels

[±round]

[-round]	[+round]
ʌ	o
ɑ	ɔ

[±ATR] or **[±tense]**

[+ATR]	[-ATR]
i	ɪ
e	ɛ
u	ʊ

Feature Theories

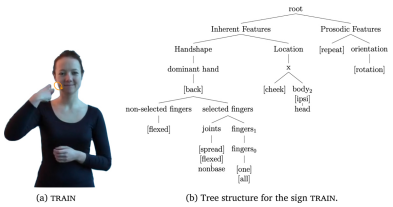
Sometimes, the feature set we have isn't quite adequate.

In Philadelphia, /æ/ gets tensed before the following segments in red:

p		t	k
b		d	g
f	θ	s	ʃ
v	ð	z	ʒ
m		n	ŋ

Feature Theories

Signed languages also have distinctive features



(a) TRAIN (b) Tree structure for the sign TRAIN.

Figure 1: "TRAIN" in French Sign Language (Mertz et al., 2022)

Using Distinctive Features

Turkish Vowel Harmony

translation	nominative	accusative
wind	jel	jel-i
tooth	diʃ	diʃ-i
rose	gyl	gyl-y
sea	gøl	gøl-y
girl	kɯz	kɯz-ɯ
branch	dɑl	dɑl-ɯ
arm	kol	kol-u

Using Distinctive Features

Turkish Vowel Harmony

- Accusative morpheme: [+high]

- Assimilates to the preceding vowel for [±back] and [±round]

References

Bibliography

Mertz, J., Annucci, C., Aristodemo, V., Giustolisi, B., Gras, D., Turco, G., Geraci, C., & Donati, C. (2022). Measuring sign complexity: Comparing a model-driven and an error-driven approach. *Laboratory Phonology*, *24*(1). https://doi.org/10.16995/labphon.6439